Assignment Coversheet – GROUP ASSIGNMENT

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Assignment Details:

Assignment Title	**Assignment 2 – Group Report**				
Assignment number	**2**				
Unit of Study Tutor	**Behroz Newaz Khan**				
Group or Tutorial ID	**CC-12**				
Due Date	**4 – Nov - 2024**	Submission Date	**4 – Nov - 2024**	Page Count	**8**

Membership

Group Name/Number	CC-12-G01				
Family Name	Given Name (s)	Student Number (SID)	Unikey	Contribution + percentage	Signature
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How would you visualise your data?

COMP5048/4448 Assignment 2

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CC-12
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I. Data Exploration and pre-processing

The dataset is collected from Ben Hamner on Kaggle [1], showing data about the US Presidential Election 2016. This event was a historic occasion that shocked many observers. Donald Trump prevailed in the election, contrary to popular expectations that Hillary Clinton would win handily. Both candidates used social media to interact with voters and sway public opinion, especially Twitter. This dataset offers a thorough compilation of tweets from both candidates during the election season.
We utilized some NLP techniques to clean the text such as removing URLs, stopwords (such as is, and, are, etc.), meaningless words, punctuation, etc. This step was to avoid the value for these words override the visuality of the other significant words [2].

II. Task A – Thematic analysis and engagement

A. Visualization Process Overview

In this part, we first find ways to identify different themes of discussion, then visually represent different themes of discussion. After that, we display the level of engagement (retweets and likes) for each theme within the same visualization.

B. First Visualization: Explore Words and Engagement

Before doing this visualization, we integrated all text, clean it as mentioned above and find the sum retweet_count and favorite_count for each word to find the engagement when that word appears in the sentence.

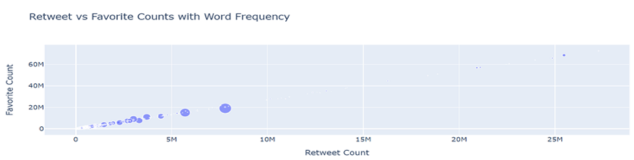


Fig. 1. Retweet vs Favorite Counts with Word Frequency Visualization

This visualization uses scatter plot. The choice of scatter plot helps showing the spreading of each datapoint, in this case, word, more clearly. It also helps find patterns in data. The axes including favorite count and retweet count are used because this visualization chart focuses on the popularity of each word; these two attributes need to be clearly displayed. This can firstly help users clearly see the distribution of words in "favorite count" and "retweet count" and secondly help users more easily identify words with more "favorite count" and "retweet count" so as to select meaningful and popular words. In this way, keywords can be found.
A certain visual variable called size is used. The size of a data point represents the number of times the word was mentioned in all tweets. The size of the data point clearly distinguishes the frequency of word occurrence, making it easier for users to understand the frequency of occurrence of specific words.
Certain interactions include zooming in and out of images, moving images, and displaying information about a data point when the mouse is hovering over it. It would assist analytic processes because zooming in and out of images and moving images help users understand the overall distribution of the data and the details they want to focus on. When the mouse is hovering over a specific data point, the data point's favorite count, retweet count, count and content of the word is shown, helping users to identify the words they want.

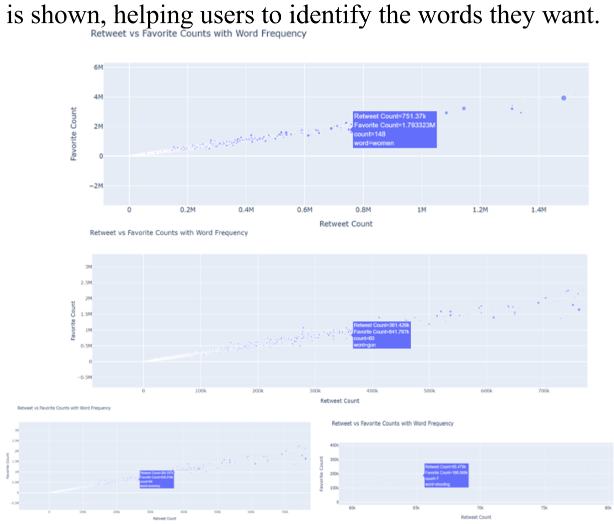


Fig. 2. Retweet vs Favorite Counts with Word Frequency Visualization with Hover Interaction

We can identify a series of words with high occurrence rates and classify them into the same theme, for example, the following figures show 4 found keywords: women in 'Women Right', 'gun' in 'Crime', 'economy' in 'Economist', 'shooting' in 'Crime'.
In the same way, we scanned through the visualization, and got a whole set of topics with keywords, for example, 'Election' includes 'president', 'vote', 'poll', 'campaign', 'Business' includes 'business', 'company', 'full-time', 'wage', 'jobs', 'wealth', 'foundation', 'charitable', 'charity', 'work', 'job', 'pays', 'manufacturing', 'outsourcing. 'Income Includes 'income', 'incomes', 'Crime includes 'crime',

'criminal', 'shooting', 'shoot', 'attack', 'violence', 'violent', 'murders', 'prisoners', etc.

C. Second Visualization: Explore Themes and Engagement By Word

After getting the list of keywords and identified themes from the 1st visualization, we decided to plot the words again but with different themes

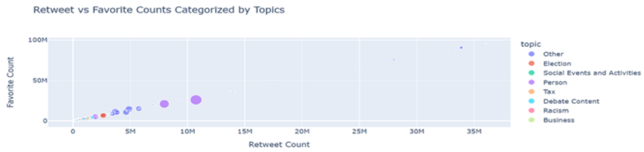


Fig. 3. Retweet vs Favorite Counts with Themes Visualization

This visualization uses scatter plot. The choice of scatter plot helps showing the spreading of each datapoint, in this case, word, more clearly. The axes including favorite count and retweet count are used because this visualization also shows the popularity of each word, like the last one. This can help users to see distribution of data points on these two attributes clearly, seeing each word's favorite count and retweet count. Certain visual variables including color of datapoints, size of data points were used, the color of datapoints shows the themes of each word, helping users understand which theme is more popular and is more mentioned. They would help users to identify which theme's words are mentioned more with "count" attribute, and which of them have more favorite count and retweet count.

Certain interactions include filtering by theme, zooming in and out of images, moving images, and displaying information about a data point when the mouse is hovering over it. It would assist analytic processes because zooming in and out of images and moving images help users understand the overall distribution of the data and the details they want to focus on. When the mouse is hovering over a specific data point, the data point's topic, retweet count, favorite count, count and word are shown.

D. Third Visualization: Different Themes with Engagement Level

This is the additional visualization that summarize the existing themes with their engagement. First, we process the data by calculating the average values: all favorite counts of each tweet in each theme are summed up and calculate the average to get the "average favorite count", the retweet count is calculated into "average retweet count" of each theme by the same method. The generated visualization shows the average value of favorite count and retweet count for each theme, with attributes including "theme", "average favorite count", "average retweet count".

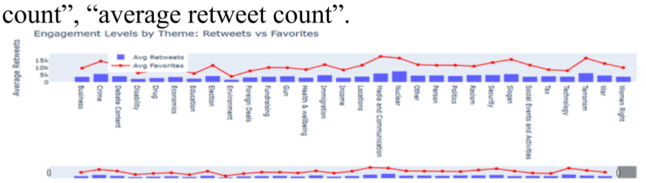


Fig. 4. Engagement Levels by Theme: Retweets vs Favorites Visualization

This visualization uses line charts and bar charts. The line charts represent average favorites for each theme, and bar charts represent average retweets for each theme. The line chart and the bar chart clearly differentiate the retweets and favorites. First, they make it easier for users to compare their changes within one theme. Secondly, bar charts and line charts help see the difference and changes between themes. Axes including values and themes are used because this visualization depicts retweets and favorites for each theme directly helps users see the average retweet and favorites of each theme more clearly. By showing the "average favorite count" and "average retweet count" of each theme, we can show the level of engagement of each theme.

Certain interactions include sliders, zooming in and out of images, moving images, and displaying information about a data point when the mouse is hovering over it. It would assist analytic processes because for slides, we can zoom in on the theme more conveniently. For mouse hovering, when a mouse is hovering over a specific theme, the theme's name, average retweets and average favorites are shown. The zooming in and out of images and moving images are of the same effect as that of the previous visualization.

E. Conclusion

In conclusion, for the visualization in Fig.1 and Fig.3 in part A, we used visualization tools to help discover and identify different discussion topics. For using evidence to define groups, as nominal data in visualization is complicated**,** we found some evidence to help us define the groups, that is, to find keywords and group similar topics together [3]. This evidence is presented through our visualizations Fig.1 and Fig.3, including patterns, trends, and themes in the data. Fig.1 shows the trend of the data in favorite count and retweet count, which basically shows a linear increase, which means that the higher the level of engagement of the data points in the upper right corner, the more priority should be given to the words. In Fig.3, after the themes are separated, the data points are colored as the classification of the themes, so that it is easier to see which themes have a higher level of engagement. Through these two figures, we completed the discovery and explanation of the grouping.

The visualization in Fig.4 visually represents different themes of discussion in a visualization. In this visualization, for each theme, level of engagement, which are represented with average retweets and likes are shown. It can be seen tha Crime, Media and communication and Terrorism are the themes with higher levels of engagement. For example, as is shown in the figure, Crime theme has 5742754 average retweets and 14.52k average favorites. Disability, Education and Environment are the themes with lower levels of engagement. For visualization design and interpretation, our visualization matches the task requirements. Our visualization clearly shows the distribution and engagement of the topic. The reason and intention behind this design is that the line chart and bar chart represent the average retweets and average favorites count respectively, so that we can see the difference between them in different topics and compare their relationship in the same topic (it can be observed that the average retweets and average favorites count basically increase and decrease at the same time) [4].

III. TASK B – SENTIMENT ANALYSIS

A. Visualization Process Overview

Through the word visualization in task A, we also obtained some emotional words and phrases. Then, we generated visualizations which explained how each candidate's theme of discussion and their sentiment evolved throughout the campaign and the tone and emotional engagement of both candidates, with themes linked to sentiment.

B. First Visualization: Explore Themes and Sentiment keywords

Derived from task A, we spotted some keywords that can show the sentiment (positive, negative, neutral). Therefore, we continue from visualization in task A, we plotted the keywords showing the sentiment per theme.

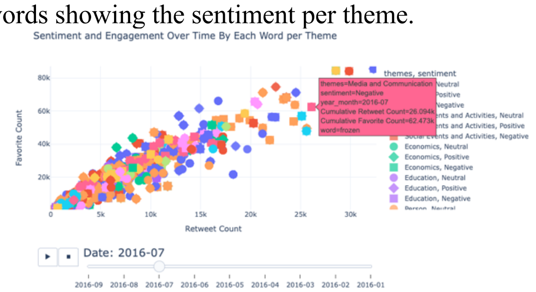
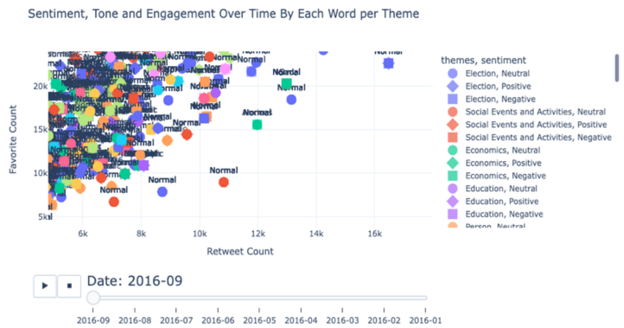


Fig. 5. Sentiment and Engagement Over Time By Each Word per Theme Visualization

The visualization uses a plot chart. The axes include favorite count and retweet count. It directly shows the audience's engagement with the words.
Certain visual variables including color of the data point and shape of the data point. The color represents the theme of each datapoint, which represents a word. The shape represents the sentiment of the data point. The color and the shape can show each word's theme and sentiment with a clear visualization, helping users to more easily analyze the data.
Certain interactions include a time slide, zooming in and out of images, moving images, and displaying information about a data point when the mouse is hovering over it. It would assist analytic processes because for the time slide, it shows the world's changes throughout the campaign. When the mouse is hovering over a specific data point, the data point's theme, sentiment, retweet count (of the row the word is in), favorite count (of the row the word is in), and word are shown. The zooming in and out of images and moving images are of the same effect as that of the previous visualization.

C. Second Visualization: Explore Themes, Sentiment, and Tone keywords

From the first visualization, in each sentiment class, we can spot some keywords can show tone such as joyful (positive), anxious, negative, etc.



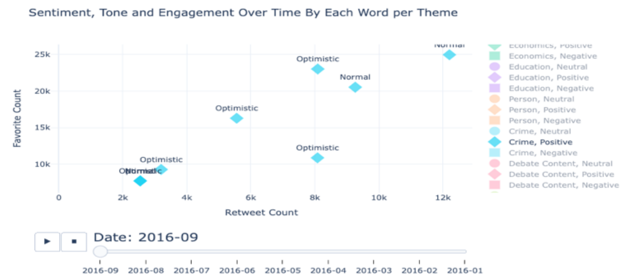


Fig. 6. Sentiment, Tone and Engagement Over Time By Each Word per Theme Visualization (with Filter Interaction)

Scatter plot is used because it is easier to see the distribution of data points, in this case, the popularity of each keyword. The axes include favorite count and retweet count. These are 2 values taken from the tweet contain the word, so that 2 words can be at the same position but different theme, sentiment, and tone. Also, therefore, the word can be duplicated because it appeared in different time and different theme. By using this, we can visualize how each word engagement difference when they are in different context.
Certain visual variables include datapoint color and shape. The color of the datapoint represents the different topics of the tweet, and the shape of the datapoint represents the sentiment of the tweet. This makes it easier for users to filter the chart and find the information they need and easier to identify different types of tweets, understand which topics or tones of tweets are more popular, and how their popularity changes over time.
Certain interactions include time slides, zooming in and out of images, moving images, and displaying information about a data point when the mouse is hovering over it. It would assist analytic processes because for the time slide, it helps to show how tweet changes throughout the campaign. The mouse hovering over a specific data point shows the data point's theme, sentiment, retweet count, favorite count, tone, time and text. The zooming in and out of images and moving images are of the same effect as that of the previous visualization.

D. Third Visualization: Explore Themes, Sentiment, and Tone of Tweets between Candidate

After getting the tone and sentiment keywords, we visualize how each tweet sentiment and tone of 2 candidates evolve throughout the campaign

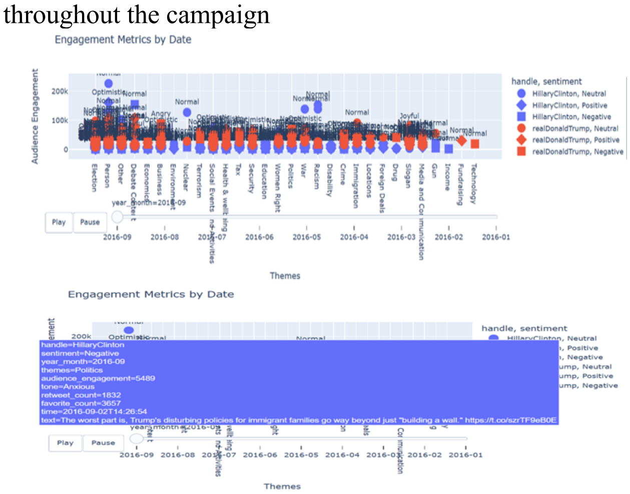


Fig. 7. Engagement Metrics by Date per Candidate Visualization (with Hover Interaction)

Scatter plot is used to show how data points of a tweet are spread on the theme and audience engagement with respect to different times. The axes let users directly see audience engagement and different themes. In this way the change of audience engagement of a certain theme throughout the campaign can be shown clearly.

Certain visual variables including color of data point and shape of datapoint were used, the color of data point indicates tweets from Hillary or Trump, to help users filter the different candidate's tweets to better understand and compare audience engagement of their tweet's. The shape indicates sentiment of "positive", "negative" or "neutral", letting user identify the different sentiment or filter the wanted tweets with certain sentiment to better analyze the data.

Certain interactions include a time slide, zooming in and out of images, moving images, and displaying information about a data point when the mouse is hovering over it. It would assist analytic processes because for the time slide, it helps show how the tweets change throughout the campaign. For the mouse hovering over data points, the data point's candidate, sentiment, theme, audience engagement, tone, retweet count, favorite count, time and text are shown, allowing users to have a more detailed understanding of each data point. The zooming in and out of images and moving images are of the same effect as that of the previous visualization.

Fig. 8. Tone Distribution over Time by Candidate with Audience Engagement Visualization (with Hover Interaction)

Another similar visualization is produced, but this is to view the change through each day so that the date is not shortened to month only. The reason why using scatter plot is the same as the previous one. However, the axes are different. They include audience engagement (favorite count and retweet count) and tone with candidates. It directly shows the audience's engagement with the different candidates and their tones.

E. *Fourth and Fifth Visualization: Change In each Theme Sentiment and Tone*

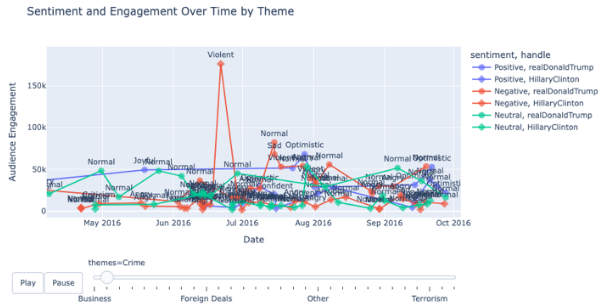


Fig. 9. Sentiment and Engagement Over Time By Theme Visualization

The line chart above visualizes how audience engagement on different topics changes over time. Existing techniques use line graphs to show the distribution of emotions, and these techniques are well suited to the time trends discussed above, proving their effectiveness [5]. The topic selection is implemented by a slide, and each data point in the line chart is the audience engagement of a tweet. The line chart has three colors, representing three sentiments, and the data point has two shapes, representing two candidates.

The visualization uses a line chart. The axes include audience engagement (favorite count and retweet count) and time by month. It directly shows for each theme, how the audience's engagement changes with the different candidates and their sentiments.

Certain visual variables including color of the data point (with the line) and shape of the data point. The color represents the sentiment of each datapoint (a tweet). The shape represents the candidate. The color and the shape of the data points can show each tweet's sentiment and candidate clearly, helping users to spot the sentiment change and more popular themes at different time periods.

Certain interactions include a slide for theme, zooming in and out of images, moving images, and displaying information about a data point when the mouse is hovering over it. It would assist analytic processes because for the slide for theme, it shows the tweet's changes throughout the campaign for each theme separately, making it easier to see details for one theme. When the mouse is hovering over a specific data point, the data point's sentiment, candidate, theme, time, audience engagement, tone, retweet count, favorite count and text are shown. The zooming in and out of images and moving images are of the same effect as that of the previous visualization.

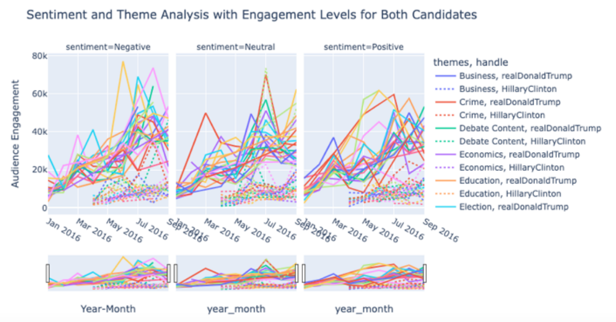


Fig. 10. Sentiment and Theme Analysis with Engagement Levels for Both Candidates Visualization

This graph illustrates how the topics and sentiment patterns for both candidates changed throughout the campaign, as well as how audience engagement and sentiment were distributed for each theme. The graph's average audience engagement for each month illustrates how appealing the various themes are, the line style sets the contenders apart, and the colors denote the subjects of conversation. This graphic aids in comprehending how the audience's reaction and each candidate's sentiment changed over time. Sentiment is depicted in this visualization in the following ways: Different colors are used to distinguish between the three sentiment categories (positive, neutral, and negative). One color is used for "Positive," another for "Neutral," and a third for "Negative." We can easily recognize lines for various mood groups thanks to this color labeling. Candidates are distinguished by their line styles: It is simpler to examine trends in audience engagement among candidates in the same mood since candidate line styles (such as dashed and solid) distinguish each candidate's performance in various mood categories. Audience Engagement: Lines are a great way to show how audiences engagement levels for mood categories change over time. The lines' placement indicates the degree of audience participation for various moods, making it easier to determine which moods elicit greater levels of audience participation. This average is determined using data aggregation's grouped averaging method.

F. Conclusion

In conclusion, in task B, we used visualization to analyze trends of sentiment changes and topic changes with timeslide.Through features such as dynamic rotation and range selection, users can analyze data more flexibly and discover potential relationships and important information [6]. The visualization in Fig. 5 shows the level of engagement of words with different topics and different sentiments at different times. The visualization in Fig.6 shows the level of engagement of words with different topics and different sentiments at different times. The visualization in Fig.7 shows the level of engagement of tweets with different sentiments, tones, and topics at different times (monthly). The visualization in Fig.8 shows the level of engagement of tweets with different sentiments, tones, and topics at different times (daily), creating visualizations for multivariable data with the scope of time [7].

For design and explain visualizations, to show the changing trends of candidate sentiment and topics. We chose Fig.7 and Fig. 8 because they show the trend changes of sentiment and topics respectively. Fig.7 shows all topics and gives the viewer an overall understanding of the month, while Fig.8 only shows the topic of the day and focuses on showing the details of each day.

For visual evidence of sentiment changes, our visualization clearly shows the changes in candidate sentiment, including the overall trend of sentiment, such as changes in positive, negative or neutral sentiment. Visualizations intuitively show how sentiment changes over time or events. Fig.7 and Fig. 8 show that most of the time is positive sentiment. In visualization in Fig.7, Fig.8, we can see that Trump is opposed to Hillary and her social events and has obtained a high level of engagement. For example, Trump's tweet on Fig.8 obtained a level of engagement of 119k. In addition, Trump often has a negative attitude towards the media. Many negative tweets were posted on June 19

For visual evidence of topic changes, similarly, our visualization clearly shows how topics change. For example, the amount of discussion or popularity of different topics changes over time. Fig.6 Sentiment and Engagement Over Time by Each Word per Theme shows that the level of engagement gradually increases over time, reaching its highest level in May, June, and July, and then slowly declines.

For comprehensive display of sentiment and theme, Fig. 7 and Fig. 8 show the changes in sentiment and theme at the same time in the same visualization chart. This can help observe the relationship between the two, such as how a certain topic changes in sentiment, which can be clearly understood through Fig.7. Or whether the change in sentiment is related to the change in theme, Fig.8 can show the daily changes, as described above for the topic and sentiment of Trump's tweets.

IV. Task C – Crital Moment Analysis

A. Hypothesis

Fig. 11. Screenshots of Critical Moment from Task B Visualizations

By analyzing the visualization in Task B, we noticed that Hillary's tweets about crime topics saw a significant increase in audience engagement between June and July 2016 for tweets containing positive sentiment keywords. Specifically, we looked at the data points for different sentiments (positive, neutral, negative) in the line chart and found that the line for positive sentiment was trending upward over that time period. At the same time, we deeply analyzed these high-engagement tweets and found that they contained positive keywords such as 'protection', 'peace', and 'sacred'. This suggests that tweets with positive sentiments received more attention during this period compared to the previous trend of tweets with negative sentiments receiving more engagement. Before that, the negative tone tweets have more engagement than the positive and neutral ones. The first positive tweet that marked the increased engagement is 'I mourn for the officers shot while doing their sacred duty to protect peaceful protesters, for their families & all who serve with them.' Although this tweet expresses the mourning, it praises the contribution of the officers with more positive keywords such as protect, peaceful, sacred so that it tends to have more positive sentiment. Therefore, we come up with a hypothetic question:

What if Hillary gave a tweet with the same content but with more negative words (showing more negative sentiment)?

B. Experiment Method

In order to test for the hypothesis, we have come up with a new alternative tweet with the same content but with more negative keywords. The new tweet is:

'***I mourn for the officers sad lost, brutal shot while trying to protect misguide protesters; this violent attack reveals society profound failure, leaving shattered families in deep despair.***'

The idea for this alternative tweet was from the below visualization. We analyzed Hillary's previous negative tweets on crime topics, especially those with high audience engagement, and extracted negative keywords that appeared frequently and were associated with high engagement. For example, we found that words such as "tragic", "violent", "failure", and "desperate" appeared frequently in these tweets. Based on this, we added as many of these negative keywords as possible to the newly conceived alternative tweets, aiming to simulate a tweet with more negative sentiment to test our hypothesis. We would like to include as many of these words as possible in order to increase the predicted audience engagement of new tweet.

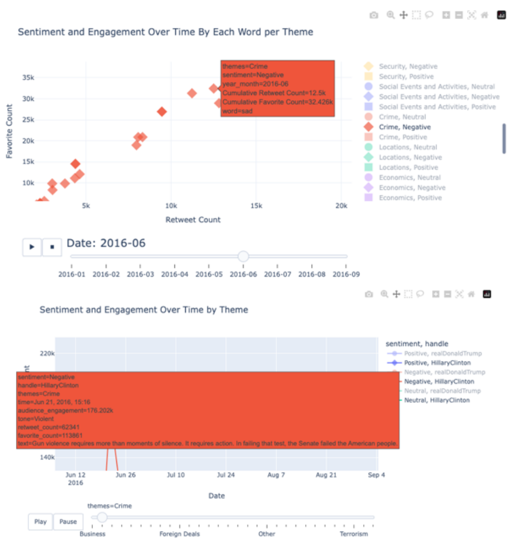


Fig. 12. Negative Keywords of Crime Theme (screenshots from task B visualization)

Specifically, we first preprocess the tweet text to remove stop words, punctuation marks, and meaningless words, retaining only keywords with actual meaning. Then, we calculate the average audience engagement corresponding to each word in all tweets. This is obtained by counting the engagement of tweets containing the word and averaging it. Next, for each tweet, we calculate the average audience engagement for all keywords it contains as the predicted engagement for that tweet. We choose to use the average method to avoid bias caused by different tweet lengths and prevent tweets from being predicted to have higher engagement simply because they contain more words [8]. In this way, we ensure that the predicted engagement between different tweets is comparable.

C. Experiment Result

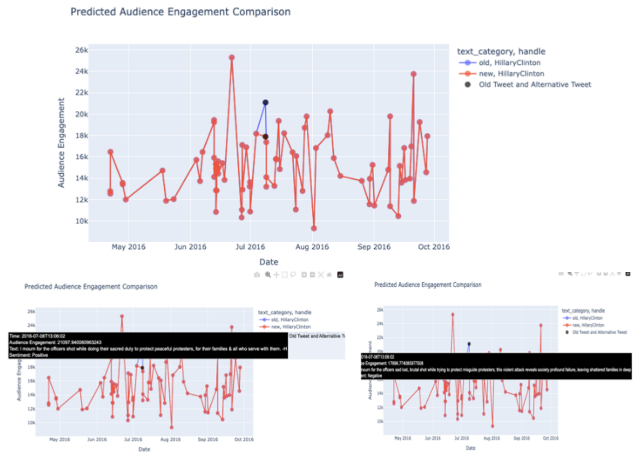


Fig. 13. Predicted Audience Engagement Change Comparison Visualization

This visualization is used to see the change in predicted audience engagement of tweets from Hillary Clinton and compare the change between the old and new tweet. Therefore, to visualize the change, we included the time of each tweet, the predicted audience engagement (because it's the main metric to compare), the text_category (to distinguish between new tweet and old tweet). Since it's the temporal analytics, using a line chart with x-axis is the time and y-axis is the predicted audience engagement is the most appropriate method. Because the text_category is the binary categorical data so we used the different hue color visual variable to distinguish between 2 categories. In order to spot 2 compared tweets, highlight interaction is used to highlight these 2 for users to easily identify which points are being compared. The interactive visualization is used in order to go deeper in details for analytical purposes. We used spanning, zooming, and hovering in this visualization because there is overlap between tweets with similar positions so that using zoom could help see those points separately. Moreover, hovering helps read the actual values of each point and more details about it to make a connection between the other details with the attributes visualized by visual variables.

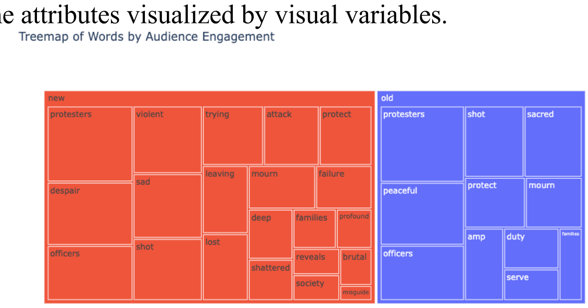


Fig. 14. Treemap of Words by Predicted Audience Engagement Visualization

This visualization is used to visualize the average engagement of each word in 2 compared tweets (old content and new content). There is the hierarchical relationship between tweet category and the words in the tweets because the words form the sentences. Therefore, using treemap is possible here. The size visual variable is used to represent the predicted average audience engagement of each word because this is the ratio data that is better to be compared by size of each word block. The color visual variable is used to represent the categories of the tweets (old or new tweet). The hover user interaction is used to show more details about each word such as average engagement count, word, text category, etc.

D. Conclusion

As shown in the line chart, the blue line is the change with the old tweet. The previous one and the one after the target tweet are the negative ones, which have lower engagement than the target one. The red line is the new tweet. Since the text other than target tweet are kept so that 2 lines are overlapped except for the target tweet. We can see the new tweet has lower predicted audience engagement than the old one. In the tree map, the predicted total audience engagement of the new tweet is more than the old tweet; however, when moving through each word with hovering interaction, we've seen that most of the words have lower average audience engagement than the old tweet. Therefore, we supposed that since the new tweet has longer length than the old tweet so that the total value is bigger. After considering 2 visualizations, we made a conclusion that if Hillary posted with a more negative tone in that critical moment, the engagement will probably not be as high as the old positive tone.

V. Evaluation

A. Set Up

Fig.1 is used to do the evaluation with other people to see if any improvement can be made. Since this is interactive visualization, it's more appropriate to conduct an interview with users to gather information. This method helps to discover facts and explore more explanations from users. In order to not cause users to have irrelevant evaluation about the plot, we decided to use a structured interview which contains a predefined set of questions [9].

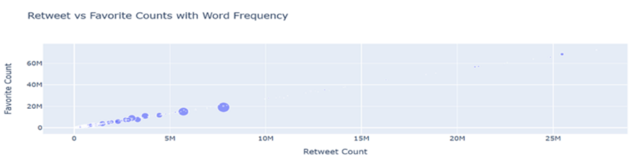


Fig. 15. Chosen visualization for Evaluation

The question list are as below:

Close-ended Questions

Question 1: Were you able to draw the main conclusions we wanted to convey through this visualization?

Question 2: Can you identify which keywords are most important in this visualization?

Question 3: Can you tell from the visualization which word has more audience engagement on social media?

Question 4: Can you understand what each data point size represents? For example, do they effectively indicate word frequency or engagement?

Open-ended Questions

Question 5: What difficulties did you encounter in understanding this visualization? What are some suggestions for improvement?

Question 6: On the whole, are these visualizations clear and intuitive, and are they effective in helping you understand the data? Please explain what you think.

This set of question contains both close-ended questions and open-ended questions. For close-ended questions, we expect the user to answer either Yes or No. Because this is an interview, we may ask user to clarify more about their choice. For open-ended questions, we allowed user to give any answer they can think of.

B. Interview Conduct

The participant with the interview was a student who is currently taking COMP5048 – Visual Analytics. With this participant, they have the basic knowledge about interactive visualization and data visualization. Therefore, it was assumed that there was no need to instruct the user basic step to do the interview and how to use the interactive visualization.

Below is their recorded response:

TABLE I. PARTICIPANT EVALUATION ANSWER

Question Number	Participant Answer
1	Yes, the visualization uses different visual variables to show audience engagement and keyword. I can see the popularity of keyword.
2	Yes, the visualization shows the most important keywords by their high frequency and the degree of interaction (likes, reposts). However, I cannot see each keyword clearly because I cannot zoom in and out or click.
3	Yes, through visual variables such as scatter plots and colors in the figure, I can intuitively see which keywords have received more attention and higher interaction volume.
4	I can understand.The size of the data point represents the word frequency. The size of the data point in the scatter plot represents the frequency of occurrence of the keyword in the tweet. The larger the size of the data point, the more times the keyword appears.
5	I think some data points overlap too much, making it difficult to distinguish different data points. More interactive features can be introduced, such as hovering the mouse to display more detailed information. Also, some data points are too small that I can't identify with bare eyes.
6	Overall, I find this visualization relatively clear, but some aspects, such as excessively overlapping data points, are not very clear and prominent, which affects the intuitiveness.

Based on the response from participant, we can come up with these points:

- Overlapped and small points: There are some small points that hard to identify with bare eyes. This may because the difference with size of each point (the word frequency)
- Unclear interaction: There is zooming interaction; however, it may cause the user a little hard to identify where to do the zooming. This might need more instruction from the group member.
- Good information conveyed: With chosen values on 2 axes and visual variables, the user can understand the main idea we want to convey through the visualization.

C. Improvement

In order to improve the readability of the data points, we decided to add the word as text visual variable on top of each data points. Also, as mentioned by the participation, there is an overlap on each data point, we decided to move the text to the right with smaller font size to reduce the overlapping. For the unclear interaction, we interviewed again with participant and gave a clearer instruction on how they could do with the visualization interaction.

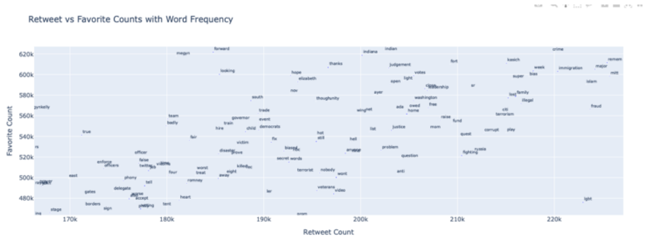


Fig. 16. Improved Visualization

The above visualization is the improved one. Here is the new response from the participant.

TABLE II. PARTICIPANT EVALUATION ANSWER ON IMPROVED VISUALIZATION

Question Number	Participant Answer
1	Yes, the visualization uses different visual variables to show audience engagement and keyword. I can see the popularity of keyword.
2	Yes, the visualization shows the most important keywords by their high frequency and level of interaction (likes, retweets). The visualization uses a scatter plot to show the number of likes and retweets for multiple keywords. I see that important keywords like “security” and “tweets” are located in the upper right corner of the chart, indicating that they are more popular in the entire dataset and I can identify them as important keywords.
3	Yes, through the visual variables such as the scatter plot and color in the figure, I can intuitively see which keywords have received more attention and higher interaction. By observing the right and top areas of the graph, I can clearly see that keywords such as "safety", "tweets", and "anger" have higher audience engagement. These keywords have received more likes and reposts than other words. I can easily tell which words have higher audience engagement by observing the position of the data points.
4	I can understand.The size of the data point represents the word frequency. The size of the data point in the scatter plot represents the frequency of occurrence of the keyword in the tweet. The larger the size of the data point, the more times the keyword appears.
5	Now I can clearly see the data points through interaction, so I think there may not be too many problems.
6	Overall, this visualization clearly shows the relationship between likes and retweets, and uses the position of the two axes to represent the popularity of different keywords. Now in areas with more data points, through the interactive sharing function, I can clearly zoom in and hover the overlapping areas to observe the data.

ACKNOWLEDGMENT

We want to sincerely thank Masahiro Takatsuka, Behroz Newaz Khan, and teaching team for all of their help and advice during this project. Additionally, we would like to thank our friend (Anonymous) for their assistance with the evaluation process. Lastly, we would want to express our gratitude to everyone who helped to successfully complete this report.

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Assignment 2 Individual Report

COMP5048/4448 Assignment 2

Quynh Giang Le
530190754

CC-12
Behroz Newaz Khan

I. Non-visualization tasks

A. Data Processing

Based on the clarification from lecturer Masahiro Takatsuka and the suggested preprocessing step in NLP, the text in the dataset should be cleaned before plotting in the visualization. The text contains some meaningless words such as stopwords (and, is, are, etc), punctuation, website link, etc. This is for the later tasks that require visualizing the word cloud. Because there are a lot of those words, it will make the other significant words harder to find in the visualization [1].

B. Document Management

I created a shared document on Google Drive for the whole group to work on it. I decided the structure of the report with the sections such as data exploration and cleaning, visualizations in task A, visualizations in task B, visualizations in task C, and then evaluation. I also decided the order of visualizations in each task.

For the code file, I was responsible for integrating every code file from the other members and delivering the final code file to the group. Since the current group chat has a function of filtering chat history by file and also the coding is not much so I used this as the code file management place.

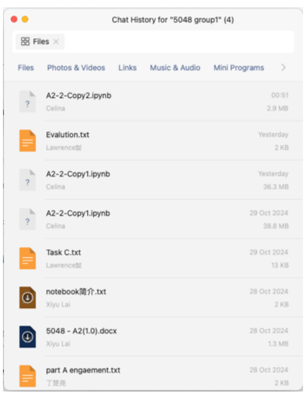


Fig. 1. File Storage of Group Chat

For the final report file, I transferred the file on Google Drive into the required template because Google Drive doesn't support two columns document so that the template would not be uploaded to Google Drive.

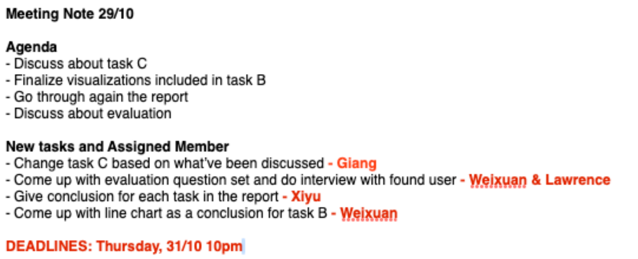
C. Programming

Throughout the process from importing data, processing it, and producing visualization, I used Python with Jupyter Notebook to complete the tasks. In Python, I used matplotlib, pandas, numpy, and Plotly libraries. I also used some NLP packages such as nltk to get a list of stopwords for preprocessing data. For the other libraries, I used them for process data for visualization. Since this assignment needs interactive visualization to complete the tasks, I used Plotly for this purpose.

Fig. 2. Sample of code for visualization

D. Project Management

For this project, I was in charge of managing the process of management. I assigned tasks for everyone in our group and managed the progress of everyone in our group. We used WeChat as the main communication group chat for daily questions. For the meeting, we used Zoom to conduct the meeting. In each meeting, we discussed the question, clarification information given on Ed discussion. We identify next steps. I summarized the discussion and defined the tasks, then I assigned them to each member. The goal was to have everyone have their tasks. Then I decided the deadlines for the assigned tasks. We agreed on the next meeting schedule. For the next meeting, we discussed what we have achieved. I integrated everything to get the final version then sent to the group for confirmation.


Meeting Note 29/10

Agenda
- Discuss about task C
- Finalize visualizations included in task B
- Go through again the report
- Discuss about evaluation

New tasks and Assigned Member
- Change task C based on what've been discussed - **Giang**
- Come up with evaluation question set and do interview with found user - **Weixuan & Lawrence**
- Give conclusion for each task in the report - **Xiyu**
- Come up with line chart as a conclusion for task B - **Weixuan**

DEADLINES: Thursday, 31/10 10pm

Fig. 3. Example of one of meeting note (recorded after each meeting)

II. Visualization Tasks

A. Task A

Based on the clarification and suggestion from the lecturer, I came up with the idea of plotting the words in the all tweet text data. At first, I tried to get the unique words list and marked the existence of each word in each row as 0 or 1. This made the dataset become a multidimensional dataset that I would apply PCA to plot the position of each point. According to the week 10 lecture, the PCA is used to visualize the high dimensional dataset by reducing to lower dimensional space. However, also in this week's lecture, it's said that PCA is hard to interpret because the position of points are different in PC1 vs. PC2 space and PC2 vs. PC3 space. Also, it's hard to include the retweet count and favorite count into the plot. Therefore, I would like to come up with a new method to visualize this.

I looked again at the dataset and decided to use interactive visualization to plot this one. I would like to include retweet count and favorite count in the same visualization apart from the word and their frequency. Therefore, I got the sum of retweet count and favorite count of every tweet that has that word. Based on the Week 3 lecture about Semiology of Graphics, I would like to visualize the correlation between retweet count and favorite count so I used an orthogonal diagram to plot these 2 values in 2 axes. Also due to this lecture content, I would like to include the word with its frequency to identify the themes because there are some specific words that can show the topics of discussion. I decided each point in the visualization represents 1 word in the list. Since the frequency is the ratio data so I decided to use the Size visual variable to visualize them to distinguish which word appeared more. However, if this is just visualized by normal visualization (static one), it's impossible to investigate each point information because there are a lot of points, so I decided to change to interactive visualization. Due to Week 4 lecture about Human Computer Interaction and Week 9 tutorial about Interactive Visualization, users can explore visualization by zooming and panning and Plotly offers this as default [2]. In order to visualize each point, users need to zoom in and zoom out to a specific location to view the points, especially the overlapped ones. Therefore, I created a plot with Plotly and include these interactions in the plot. Moreover, in order to know more details about each point, I also included the hovering interaction for the user to see the details because there are more details to look at other than the ones on the axes such as the word text, specific retweet count and favorite count, etc.

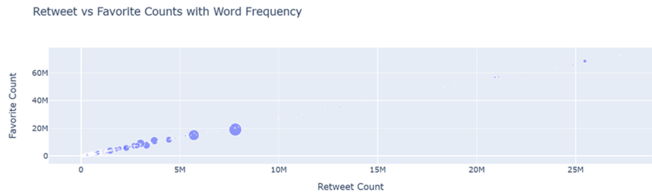


Fig. 4. Retweet vs Favorite Counts with Word Frequency Visualization

After getting the above visualization, I move around the visualization to find the keywords that define the topics. Then I grouped them into a list of topics with their keywords. In order to make it clear about the found topic, I visually created a similar visualization that defines the topic that each word belongs to. This visualization was to have a clearer visualization about the detected topics with their keywords. The visualization is as below:

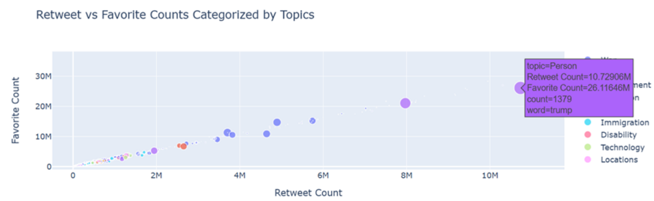


Fig. 5. Retweet vs Favorite Counts with Word Frequency with Topics Visualization

Since the topics are nominal data so that utilizing color with different hue can best represent this according to Week 3 lecture and exercises from Week 3 tutorial.

B. Task B

Based on the keyword visualization in task A, I also spotted keywords that show the sentiment and tone such as crooked, sad, sacred, appreciate, etc. Some of them sound positive, some of them sound negative, and others have neutral feelings. I spent time moving around visualization in task A to find keywords. The sentiment they gave me was as general as just positive, negative or neutral. Therefore, I grouped the keywords to these 3 sentiments and visually show them by the following visualization:

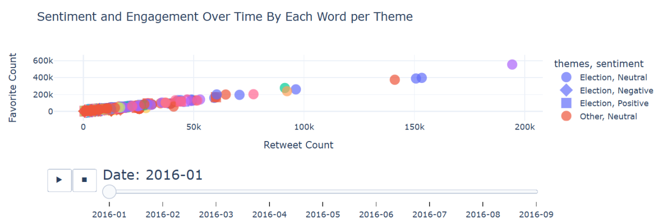


Fig. 6. Sentiment and Engagement Over Time By Each Word Per Theme Visualization

The requirement of task B asked to give the evolution of sentiment of each theme throughout the campaign so I came up with the idea of integrating the words in task A and their themes (the themes are the themes of the whole tweet that contains the word), sentiments with their keywords. Task B also asked for showing the engagement so that I included the favorite count and retweet count, similar to task A. Because of the need of showing the change so that the time attribute was included. The reason for choosing an orthogonal diagram with retweet count and favorite count on 2 axes are the same as task A. The default interaction of zooming and panning are used here, same as task A. In the Week 6 tutorial, the time attribute can be shown by the slider in interactive visualization so I used the slider here for the time. In Week 3 lecture, the categorical data can be distinguished by hue and symbol visual variables so I used both here for the theme and sentiment.

By moving around the above plot, I spot that in each sentiment, there are some keywords that show specific tones such as joyful, angry, criticize, etc. I again came up with the keywords for some specific tones in each sentiment and make an update to previous plot as below:

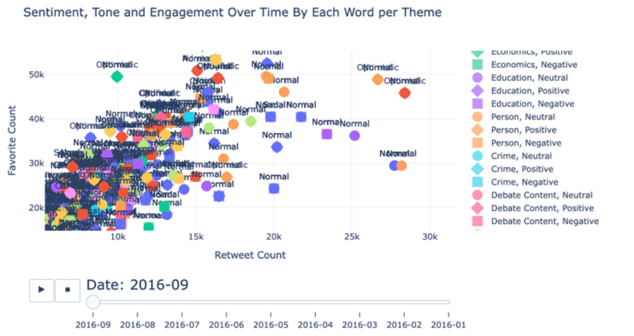


Fig. 7. Sentiment, Tone and Engagement Over Time By Each Word per Theme Visualization

Since the tone is the categorical data and we need to see it right away so that text label visual variable was chosen to present this variable.

These are just the words, I would like to visualize the tone and sentiment of the whole tweet. I came up with the following visualization

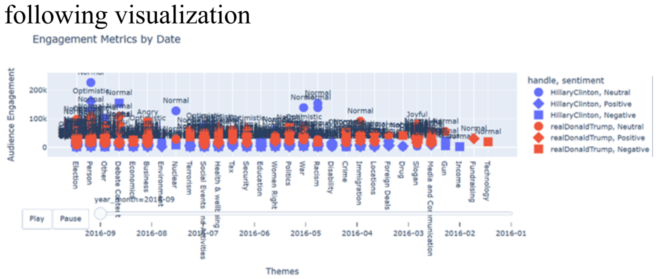


Fig. 8. Engagement Metrics By Date for each Candidate Visualization

I would like to visualize the tweets of each candidate. Also, since the retweet count and favorite count have positive correlation so that I sum them up to get the audience engagement attribute. I wanted to plot the correlation between the themes and audience engagement so that I used an orthogonal diagram (Week 3 lecture). There are only 2 candidates so that using color can help distinguish between 2 and keep the visualization as a good design by not showing too many colors (Week 8 lecture). Filter interaction can be used here by filter out by each candidate and each sentiment. As stated in the Week 7 lecture, the temporal analysis should use the line chart. I came up with the idea of producing the line chart to see the change clearer. The instruction is said to show the evolution of each theme so I changed the x-axis to time and the themes to slider to get the trend in each theme as following visualization:

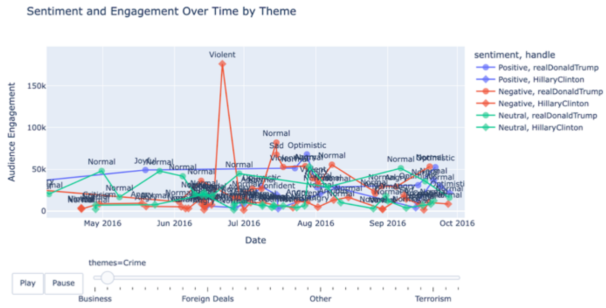


Fig. 9. Sentiment and Engagement Over Time By Theme Per Candidate

C. Task C

From visualization in task B, I spotted the increase in the positive tweet in the Crime theme of Hillary so that getting the idea from the lecture in Week 10, when talking about PCA, the lecturer had given an example of giving a hypothesis of unseen data based on the visualization, I came up with the hypothesis of what if scenario that Hillary gave negative tweet at that time. Also, referenced from Week 1 lecture, the visual analytics process is data representation & transformation -> visual representation -> analytical reasoning -> production. I did the first 2 in task A and task B, I analyzed from task B visualization to get the critical moment then made a hypothesis of alternative tweet in analytical reasoning.

Since the question asked to visually show the alternative tweet and how it might change the response from the opponent or the public and the example given by our tutor in the Week 13 tutorial session, I came up with the idea of making a visualization that compares the new tweet with the old tweet in the same visualization as below.

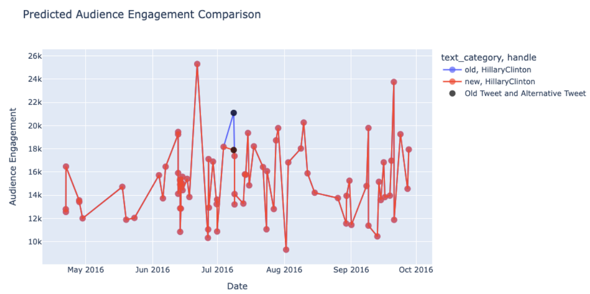


Fig. 10. Predicted Audience Engagement Comparison between Old and New Tweet Visualization

Since this is just 1 theme and I wanted to focus in the time to the evolution, I eliminated the visual variables that categorize the themes and sentiment. However, in hovering data, I included all details about text, sentiment, audience engagement, etc. for users to see all details. From Week 4 lecture, there is 1 new user interaction that was used here is the highlight. This interaction was used to highlight 2 compared tweets because it's hard for users to spot these 2 tweets based on the normal visualization. The predicted audience engagement was calculated by average because it will reduce the bias to the length of the tweet. [3]

I also wanted to investigate the predicted engagement of each word in 2 compared tweets by this treemap:

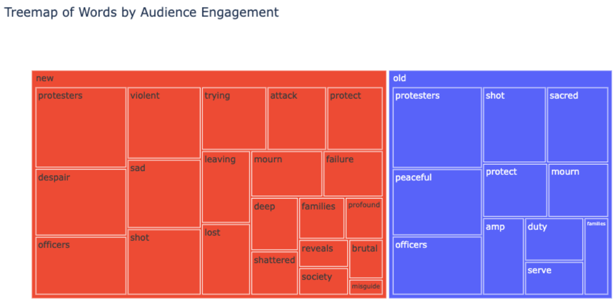


Fig. 11. Treemap of Each Word Engagement in Old and New Tweet Visualization

From the Week 7 lecture, treemap is used to visualize the hierarchical data. I think the word and tweet category (old or new) can have this relationship so that treemap was used. The color visual variable was used for the tweet category and hovering interaction was used to show more details about the engagement data of each word.

D. Evalutaion

Based on the Week 6 lecture, the interview is a method of evaluation. By using this, we can directly discover facts and opinions from users. There are 2 formats of interview: Structured and Open-ended. Since the chosen plot to do the evaluation is the interactive visualization, I think it's necessary to do the direct interview to see how they use the visualization and explore more about their feedback. I suggested the group do the structured interview with a predefined set of questions to avoid the unnecessary feedback from users.

The interview process was conducted and executed by the other teammates. After getting the response from the participant, I made the summary of critical points and made the improved visualization.

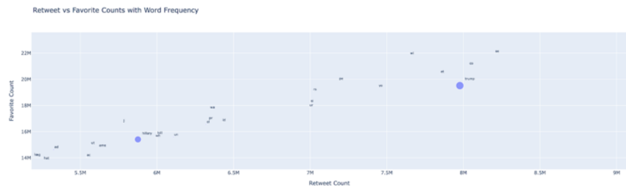


Fig. 12. Improved Visualization after Evaluation

ACKNOWLEDGMENT

I want to sincerely thank Masahiro Takatsuka, Behroz Newaz Khan, and teaching team for all of their help and advice during this project. Additionally, I would want to express our gratitude to everyone who helped to successfully complete this report.

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Inidvidual Report

COMP5048/4448 Assignment 2

line 1: Xiyu Lai
line 2: *540521184*

line 4: CC-12-G01
line 5: Behroz Newaz Khan

I. Contribution to the group wrok

This section describes the non-visualization tasks I carried out towards the group achieving its task, including:

A. Data processing,

I processed the database data with Quynh Giang Le. After getting all the text from the tweet, I visualized a bar chart based on the maximum frequency of the word. Then I added keywords with topics into a dictionary part A.

B. Document management

I wrote for most parts of reports of part A and part B, and a small part for part C.

Aiming at allowing the group to quickly understand the progress and the goals of the task, I sent several text files with brief description of the contents of the file to group chat, including word form of initial report, txt files that explained what is done in part A and part B, providing information for teammates, to ensure that our understanding of the task and current progress was as consistent as possible, and to ensure efficient communication and that everyone was working in the same direction.

C. Programming

I made a series of code attempts for tasks A and B, a dictionary including keywords and topics was generated. One visualization was used in report for part B 3.4. Visualization 4, showing the visualization for part B.

D. Deriving Visualization Application

In jupyter notebook I used plotly to process data and generate visualizations. Before choosing the final visualization to put into report, I tried several visualization with different forms and compared them before choosing the most suitable one.

E. Literature survey,

I searched and got 7 references into the report. Two for task A, two for task B, two for part C, and one for evaluation.

F. Project Management

Our group has 4 people, and are divided into 2 to finish different tasks most of the time. We split the job. First in part A, I processed the dataset to get the dictionary containing keywords and themes with Quynh Giang Le. Then I worked on part B to get the sentiment and the tone and presented all the visualization with Quynh Giang Le.

Also, I work with team mates to discuss and plan the approximate progress of the task, such as the initial plan of "5 days for Task A, 5 days for Task B, and 10 days for Task C" and before the final submission, we agreed on the estimated finish time of the individual report is 1 day before deadline and the final report 2 days before deadline.

Our group has different nationalities, so often I acted as a Chinese-English translator to help other team members understand each other during meetings or communication, aiming at improving communication efficiency, and ensuring a smooth progress of tasks.

For my different version of program files, I named my each version with a version number as suffix. Starting from 1.0, For minor changes, add a number like 1.1. For major changes, add the number such as 2.0, making it easier to distinguish.

II. Contribution to the visualization-related tasks

This section describes what I did to make contributions towards "visualization-related" tasks, then justifies my contribution with respect to what I have learnt in this unit, referencing what's been discussed in the relevant weekly contents.

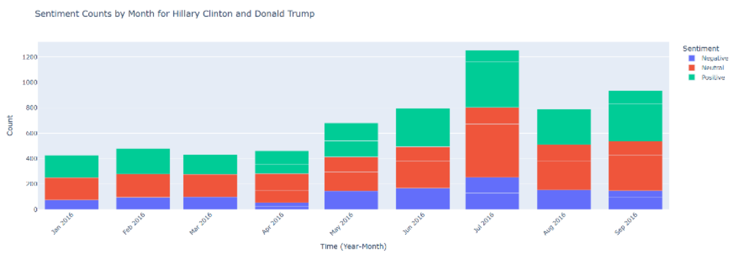
A. Determining/Designing a visualization method to be used

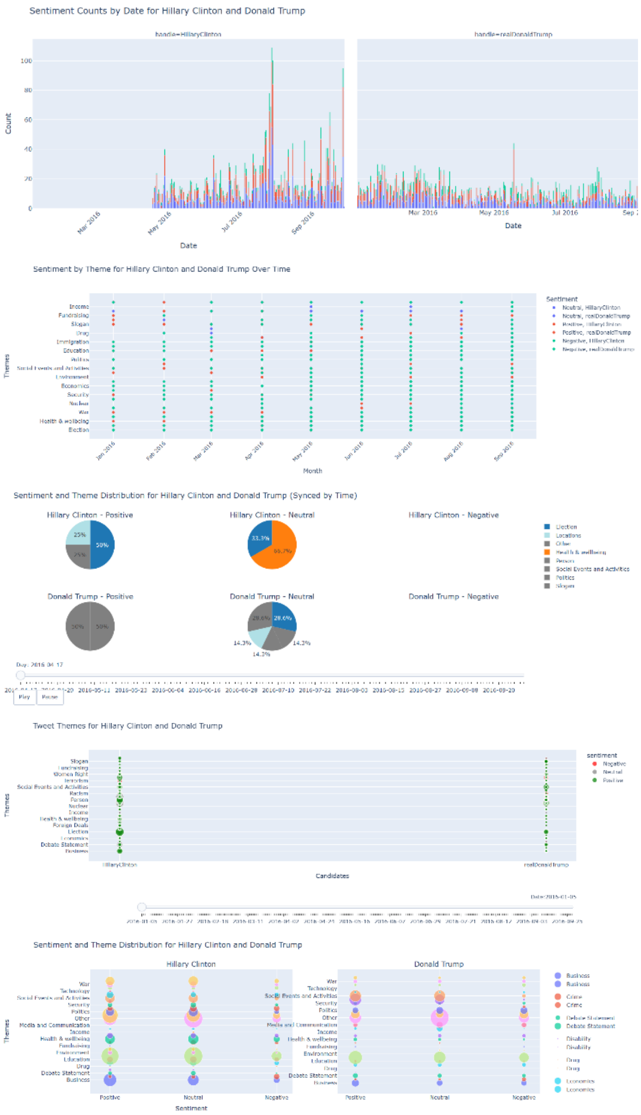
According to week 6 ppt, for the EDA, I tried to use ranking analysis to compare the frequency of each word by ranking using a bar chart.

I participated in discussion by meeting and group chat, presented our own generated charts and ideas for 3 tasks, and confirmed with team mates whether such ideas are correct.

For my visualization in part B, I decided how to create it, and shared my visualization and its updates with team mates.

I generated several attempts at part B, they helped to inspire me to choose and optimize the final version of visualization, they themselves did not explain the problem given in part B well, so I discarded them. Some of them are because the pie chart and bar chart cannot directly display the distribution, making viewer difficult to extract information from them, and some of them didn't display all the attributes required by the task. Some of the figures are shown below.





B. Determining/Designing visualization

For visualization design, I considered appropriate chart type and visual variables. I discussed with my team mate, and made a significant change in grouping words with the same theme. At first, we tried to use clustering technology such as NLP, but later, based on the ed, the teacher's questions and our discussions, Quynh Giang Le and I decided to use visualization to find keywords and then classify them by topic, rather than using programming to cluster them, and confirmed the decision to the group.

Part B was originally explained by many visualizations. Finally, based on the ed, the assignment requirements and our understanding, Quynh Giang Le and I proposed that we should use one visualization to explain all the problems in part B. So we used the previous visualization as a reference and decide that one finalized visualization will contain all the attributes and requirements of part B.

According to week 1 ppt, when designing the visualization of tweet analysis, I adopted a data-driven approach, selecting the appropriate chart type by observing the actual distribution of word frequency and sentiment, such as referring to the bar chart that shows high-frequency words. This approach ensures that my design is based on data features rather than preset visual models.

According to week 1 ppt, in the report, I used different colors and shapes to represent sentiment categories, which follows the principle of visual variables and helps users quickly distinguish between positive and negative tweets. Through color contrast, I further ensure information is distinguishable and the chart is easier to understand.

According to week 2 ppt explaining sentiment and candidate information, the sentiment and candidate information of tweets can be considered as nominal data of text categories, because this information is mainly used for labeling and classification without containing order or quantitative relationships. Temporal data are treated as interval data because they have quantifiable time intervals between different events. For data processing, in the process of data visualization, operations such as Mathematical operation on numbers (+), filtering, transforming, splitting, and merging were applied to better extract and display information.

According to week 8 ppt, my visualizations make sure to avoid exaggeration in color selection and use low-saturation colors. For keeping the consistency, the same color represents a theme. For decorative visual features, I made sure there were no extra decorations, no redundant information was displayed when the mouse hovered over the data point, no unnecessary border lines, and avoiding 3D effects. In selecting charts, I avoided pie charts and used scatter plots to show distributions.

C. Determining/Designing user interaction,

In using data points' content to convey significant information, the scaling and movement of data are built-in functions of plotly, so I used plotly to facilitate user interaction. In several visualizations, I used time slide to display information in different time.

According to week 4 ppt, the capacity of human working memory is limited, so to let users more easily extract and remember key data, I tried to simplify the information hierarchy and reduce the complexity of information in several aspects. First , I made sure information is display in steps and the interaction process is simplified. Second according to Hick's Law, I avoided presenting a large amount of data at one time, to reduce the complexity of decision-making in the process of selecting visualizations and optimized the layout of menus and buttons so that users can quickly find the main operation options and reduce decision-making time. Third, according to the "Fitz's Law", I ensured that the size of the visualization key buttons or functions is reasonable and the important interactive elements are easy to click. Fifth, according to the Ecological Model, I kept the interface at the beginning of the visualization as all the information in that frame to avoid misleading the user's focus because of some unshown data. Sixth, According to the Social Interaction Model, I uses the mouse hover to display information.

According to week 5 ppt, for data selection and goal setting, according to the GIGO principle, it is crucial to ensure data quality. Therefore, I used filtered data to visualize, identified keywords and themes at the beginning to ensure that the visualized data is appropriate or meaningful to produce effective information. For user interaction design, I used statistical comparison methods of time comparison, helping users extract valuable information from the data.

D. Determining/Designing the evaluation process,

I discussed in the group discussion. Together, we decided to find a few classmates to show them one of our visualizations for the evaluation, then we decided on the visualization image to be displayed.

After making the visualization, we collected feedback from students to ensure that the readability and effectiveness of the chart meet the needs of the target users. This is adapted to the User Feedback of week 1 ppt.

E. Conducting and analysing the evaluation

I actively participated in the group meeting as we selected the questionnaires for evaluation through group meetings. We ensured that we kept questions simple, be clear and concise.

According to week 6 ppt, we make sure we (1) group questions appropriately; (2) give explanation, keep questions simple and clear, avoiding complex terms or ambiguity; (3) provide explanation, add a brief explanation like example to the question when necessary to help participants better understand the context of the answer.

How would you visualise your data?

COMP5048/4448 Assignment 2

line 1: Lawrence Zhou
line 2: *530637662*

line 4: CC-12-G01
line 5: Behroz Newaz Khan

I. Contribution to group work

My primary responsibilities in the group project focused on Task C, specifically identifying and analyzing key moments during the 2016 U.S. presidential election. The following are my primary non-visual contributions to the group's goals:

A. Data processing: I partially processed a dataset collected from Kaggle, which contains tweets from Donald Trump and Hillary Clinton during the 2016 campaign. I provided code for some data cleaning (removing URLs, mentions, and special characters), unifying column names, and adding a "candidate" column to distinguish tweets from each candidate. These tasks were to prepare the data for my further analysis and visualization in task c.

B. Identification of key moments: Through time series analysis, I helped identify key moments in the election and analyzed spikes in tweet volume. I focused on key events, especially the increase in Donald Trump's tweets about immigration policy in September 2016. I also identified a representative and controversial tweet, further performed sentiment analysis and made suggestions for improvements, but in the end, Celina made the final decision on the selection of tweets.

C. Discussion and selection of visualization methods: I proposed a method to visualize the data of Task C in the uploaded Task C file, and discussed it with team members, and finally selected the method in the submitted file. During the discussion, we had a deviation in our understanding of Task C. In order to solve this problem, I proposed a new solution through personal thinking and combined with the professor's feedback on the ED platform, and informed the team members to ensure that the appropriate peak time was selected and reasonable predictions were made.

D. Document management: In addition to participating in the analysis, I shared Google Drive documents with the group and actively participated in writing some group documents, mainly responsible for task C, and wrote evaluation questions.

II. Contributions to visualization-related tasks

In the visualization task, I contributed to the visualization design of the dataset and discussed with the group the best way to effectively convey key information. The following are my contributions during the group's visualization related tasks:

A. Deciding on the visualization method: After analyzing the nature of the dataset, I decided to use a combination of line charts, scatter plots, and bar charts to show the changes in topics and sentiment. Scatter plots were used to highlight peaks in tweet activity and engagement because it best represents the data distribution and outliers.

B. Visualization Design: I designed several visualizations to analyze tweet content, topics, and sentiment. However, these visualizations were not used because the team members provided better ones. Specifically, I created:

1. The line overlay shows the trend of the number of tweets counted by month from the dataset over time. The horizontal axis is time (month) and the vertical axis is the number of tweets. The blue curve shows the fluctuation of the number of tweets over time, and the filled part shows the area of the number of tweets. As shown in the figure 1 below:

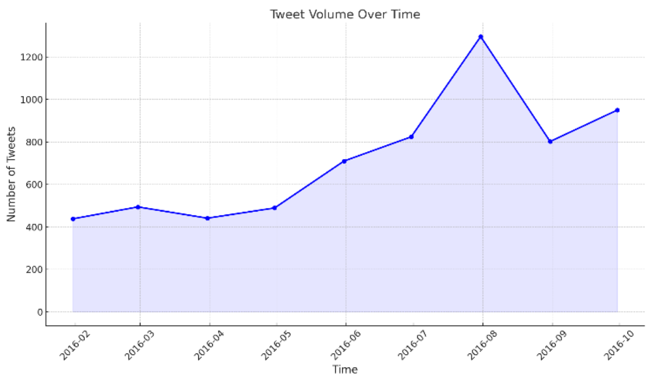


Fig. 1. Tweet Volume Over Time

Explanation: This chart shows how the number of tweets changes during the election. You can clearly see the peaks in tweet activity, which usually reflect the outbreak of some key events or hot topics. For example, peaks may be caused by candidates holding important events, debates, or making controversial remarks, which arouse more public attention and discussion. This information is very helpful for understanding the dynamics of public opinion during the election and the temporal changes in the influence of candidates.

Improvement: Add interactivity to the chart, introduce different colors or line types to distinguish different types of events or candidates, and when the user hovers over a peak of the curve, detailed information about the event, such as the event name, date, and related description, will be displayed. Sliders have been added to view different time points. The resulting chart will be more informative, making full use of the data in the file and helping to gain a deeper understanding of the dynamic changes in public opinion during the election.

2. The relationship between the number of likes and retweets of a tweet is shown in a scatter plot. The horizontal axis is the number of likes of a tweet, and the vertical axis is the number of retweets of a tweet. Each dot represents a tweet, and the transparency and size of the dot make the data distribution clearer.As shown in the figure 2 below:

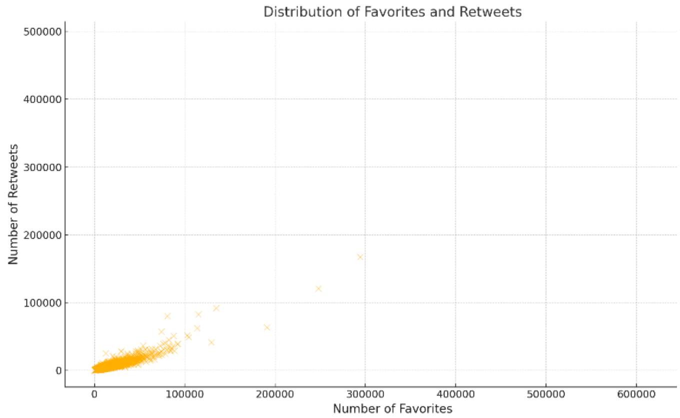


Fig. 2. Distribution of Favorites and Retweets

Explanation: This scatter plot is used to explore the relationship between the popularity of tweets (measured by the number of likes and retweets). Generally speaking, there is a certain positive correlation between the number of likes and the number of retweets, and popular tweets will have more retweets and likes. However, you can also see some outliers, such as tweets with many likes but few retweets, which may be because the tweet is an expression of personal emotion rather than content suitable for widespread dissemination. Understanding the relationship between likes and retweets can help analyze which tweets are more viral and influential.

Improvements: In the final version, we added regression lines and calculated correlation coefficients to more intuitively show the relationship between likes and retweets. Second, we used different colors and shapes to distinguish different types of tweets, such as tweets from candidates, media reports, or public comments. This helps to reveal the differences in popularity and spread of tweets from different sources. In addition, we adjusted the transparency and size of the data points to make high-density areas more obvious. We also added interactivity. When the user hovers over a data point, detailed information about the tweet is displayed, such as content summary, publishing time, and author, and a slider is added to select a time point.

3. A box plot showing the distribution of likes for Hillary Clinton and Donald Trump’s tweets. The horizontal axis shows the person who posted the tweet (Hillary or Trump), and the vertical axis shows the number of likes for the tweet. Using a log scale makes the data distribution clearer. As shown in the figure 3 below:

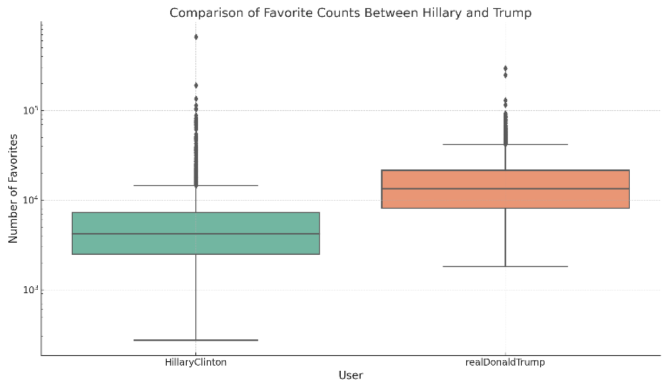


Fig. 3. Comparison of Favorite Counts Between Hillary and Trump

Explanation: This chart compares the popularity of Hillary Clinton and Trump's tweets. The box plot shows the median, interquartile range, and outliers for each candidate's tweets. By comparing the number of likes for the two candidates, you can see who has more influence on social media. For example, if one candidate has a higher median and interquartile range, it means that the candidate's tweets resonate more with the audience. The logarithmic scale is used because the range of likes is larger, which can better show the overall data distribution and reduce the misleading caused by extreme values.

Improvements: We used a scatter plot to show the attitude and popularity of Hillary Clinton and Donald Trump's tweets on different topics. Different data point colors are used to represent the sentiment of the two electorate's tweets (netural, positive, negative), and the horizontal axis shows different topics. Through these improvements, the chart not only shows the popularity of the two candidates' tweets, but also reveals their performance on different topics and sentiments. We also added interactive features, allowing users to hover over any data point to view detailed information, including a summary of the tweet content, publishing time, number of likes, number of reposts, etc. These improvements make full use of the data in the file, making the final generated chart more intuitive and informative, helping to deeply understand the public opinion dynamics and the influence of the two candidates on different topics during the election.

4. Another box plot shows the distribution of retweets of Hillary Clinton and Donald Trump’s tweets. The horizontal axis shows the person who posted the tweet (Hillary or Trump), and the vertical axis shows the

number of retweets of the tweet. Again, a logarithmic scale is used to show the distribution of retweets. As shown in the figure 4 below:

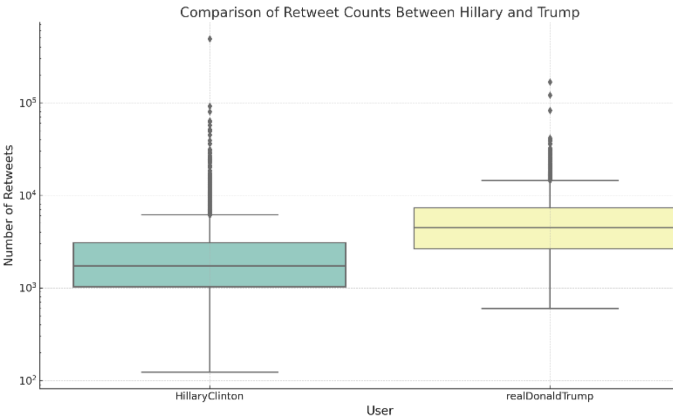


Fig. 4. Comparison of Retweet Counts Between Hillary and Trump

Explanation: The number of retweets can reflect the breadth of a tweet's spread and the degree of discussion of a topic. The boxplot compares the distribution of retweets of Hillary and Trump's tweets, which can help us understand their respective information dissemination effects on social media. For example, if Trump's median retweets and range are higher, it means his tweets are more viral and topical. Through this information, it is possible to conduct an in-depth analysis of the two candidates' interactions with the public on social media and the difference in topic influence.

Deprecated: The team decided to deprecate this box plot comparing the number of retweets of Hillary Clinton and Donald Trump's tweets. While the chart provides a basic overview of the reach of the two candidates' tweets, it does not reveal the differences and trends in their interactions on specific areas or topics. The chart lacks interactivity and does not allow users to drill down to specific data points, such as the content of the tweets, the time of posting, and the sentiment. In addition, the visual presentation is not intuitive enough to effectively attract the audience's attention and does not make logical sense. Therefore, we chose to use a more interactive and informative chart that can more comprehensively show the candidates' influence and public engagement on different topics and time periods to better meet the needs of the project.

These charts were basically the most basic version, but after discussion and evaluation by the team, it was found that these charts did not fully meet the project requirements. These charts failed to clearly show the differences and trends in the interactions of candidates in certain areas. Therefore, these charts were eventually improved or replaced with more interactive and visual versions.

C. Word cloud design: I generated two word cloud maps for Trump and Hillary's tweets, removing stop words, punctuation marks, and meaningless words, and only retaining keywords with actual meanings, to help us better understand the focus of different candidates' topics on social media and the main content of public discussion. Through the word cloud map, we can intuitively see which words are frequently used in the tweets of the two candidates, so as to better compare their information delivery strategies. However, the one provided by my teammates was more intuitive and convenient, so it was not adopted. As shown in the following two failed figures：

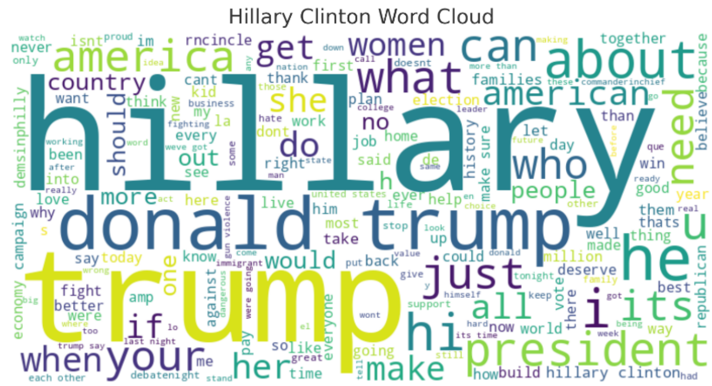


Fig. 5. Hillary Clinton Word Cloud

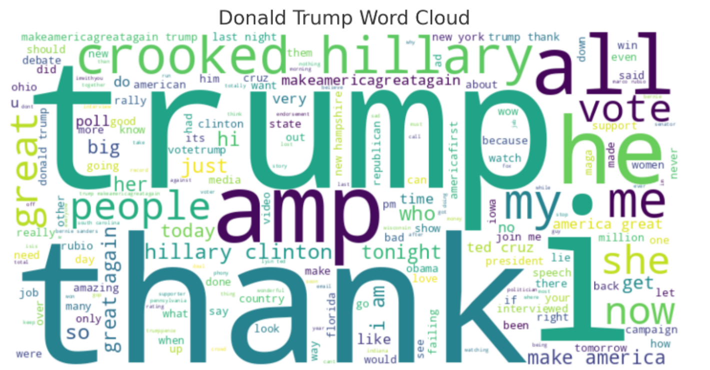


Fig. 6. Donald Trump Word Cloud

Deprecation: We chose to use the word cloud provided by celina instead of the version I originally generated. Although the word cloud I made tried to help us better understand the focus of different candidates' topics on social media and the main content of public discussion by removing stop words, punctuation and meaningless words, the word cloud map of the team member was better in intuitive comparison and emotional expression. His design is more intuitive and can clearly compare the differences in common words in Trump and Hillary's tweets. At the same time, his word cloud map is more expressive in emotional comparison, allowing the audience to more intuitively feel the difference in the information delivery strategies of the two candidates. These advantages make his chart more in line with the project needs and were eventually adopted by us.

D. Evaluation Process: I used the group's visualization to evaluate with others to find opportunities for improvement. Since this was an interactive visualization, it was more appropriate to conduct interviews with users to gather information. This approach helped to uncover facts and explore more interpretations from users. To avoid irrelevant evaluations of the visualization by users, I used structured interviews with a set of pre-defined questions and based on these feedback, I provided feedback to the group members and continued to improve the visualization.

III. Link to Weekly Course Learning

In the Week 3 course, we learned how to use data visualization techniques to display data distributions and identify outliers. I applied this knowledge to determine which type of visualization would be best for our data. For example, when analyzing tweet activity, I used scatter plots to highlight distributions and peaks in the data, helping us better identify outliers and trends in the data. In addition, I used the explanation of data distribution in Week 3 to determine the form of the graph to more effectively present the data characteristics of tweet activity.

The 5th week of the course focused on the effectiveness of user interaction and visualization design, especially the importance of interactive features for visualization analysis. These features include zooming, filtering, and mouse-over details, allowing users to explore and analyze the data more deeply based on their needs. I added these interactive functions into the visual design so that users can operate the charts more flexibly, thereby in-depth understanding of the information behind the tweet data, and improving the practicality and ease of use of data visualization.

Week 7 of the course revolved around sentiment analysis, providing the basics of conducting sentiment analysis. I applied this knowledge to analyze the sentiment scores of tweets through Text Blob and compared the sentiment changes of the original tweets with the improved tweets. Through this comparative analysis, I am able to assess whether a candidate's communication strategy on social media is effective and recommend how the content can be improved to better influence the emotions and attitudes of the audience. This application of sentiment analysis helped us understand which content was more likely to inspire positive reactions.

The course in Week 9 emphasized the importance of choosing appropriate colors and decorative visual features when designing visualizations to ensure that information is clearly and accurately conveyed to the audience. I followed this principle in my design, avoiding excessive use of color and unnecessary decoration, and ensuring that the presentation of data was concise and clear. For example, in the histogram of the sentiment score comparison, I used contrasting colors to distinguish the original tweets from the revised tweets, so that the audience can understand the change in sentiment at a glance. By reducing unnecessary visual noise, the chart can better highlight key information and improve the effectiveness of information transmission.

How would you visualise your data?

COMP5048/4448 Assignment 2

line 1: WeixuanDing
line 2: 5307975674

line 4: CC-12
line 5: Behroz Newaz Khan

I. CONTRIBUTION OF NON-VISUALISATION TASKS

During our group project work together. I focused on preparing the data and identifying themes. I also handled data processing and consolidation tasks that set the groundwork for our teams upcoming visualizations and analyses. I made contributions in these areas:

A. Data Preparation and Theme Identification :

In the data preparation phase I employed Natural Language Processing (NLP) approaches such as TF IDF and LDA for theme modeling. These techniques helped in identifying and categorizing the discussion topics within the tweet data enabling the team to concentrate on specific subjects, during further analysis tasks. This aspect of the tool is mentioned in the description of text mining methodologies [1].

B. Categorising tweets and aggregating engagement:

Once I sorted out the topics, from the tweets and grouped them accordingly I tallied up the total retweets and likes for each topic to create a comprehensive data table. This method allows us to neatly summarize and grasp the interaction levels for each topic[2].

C. Data Aggregation and Calculation of Audience Engagement by Sentiment and Candidate:

In the analysis of audience engagement, I used data aggregation to group the audience by year_month, candidate (handle), themes, and sentiment and calculate the average of audience engagement for each group. This section is referenced in the explanation of relational data and aggregation methods [3].

D. Contributions to Project Management:

In project management tasks too, I had a role to play. Chiefly in coordinating team schedules and distributing tasks efficiently while resolving conflicts that arose among team members or within the team itself. I drew upon strategies for managing teams and projects with a focus, on coordinating communication and resolving conflicts [4].

E. Solution Adjustment for Task A:

During the process of addressing Task As challenges and tasks at hand, on tweets multitopic attribution issue was inadvertently neglected by us previously, the professor's input on the ED platform was taken into account as I put forward a fresh data cleaning approach and revamped the visualization to enhance the accuracy of data representation This enhancement procedure drew inspiration from evaluation techniques and user oriented design philosophies [5].

II. CONTRIBUTION OF VISUALISATION-RELATED TASKS

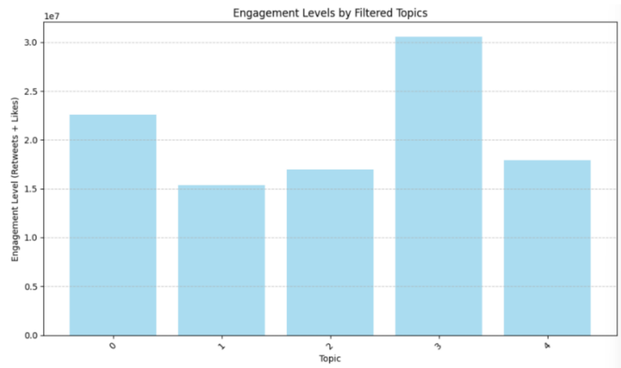
A. Identifying and designing visualisation methods:

I was responsible for designing and creating visualizations that showed the level of interactive engagement across topics. During the design process, I experimented with various combinations of charts and graphs to show engagement levels and sentiment analyses of tweets, but some of these methods failed to be adopted due to insufficient design. I will detail these failed versions and their shortcomings, as well as how they could have been improved.

B. Failed versions and shortcomings:

1) Failed Version 1 (visualization 1):

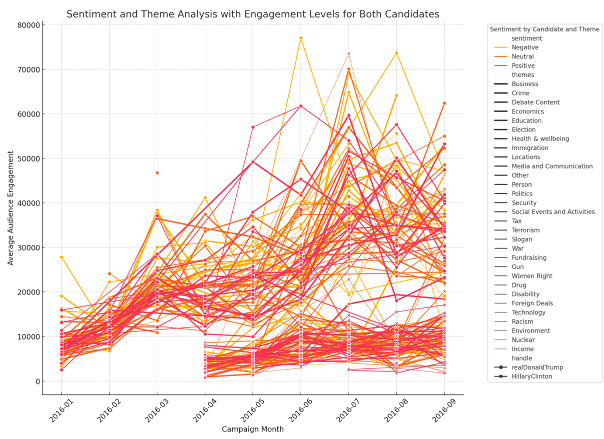
The initial design presented data using a bar graph to display interaction levels based on different topics but lacked a clear structure and failed to highlight nuanced distinctions between topics effectively. Moreover it did not consider changing trends over time which could have provided insights, into the dynamic nature of the data [6].



visualization 1

2) Failed version 2 (visualization 2):

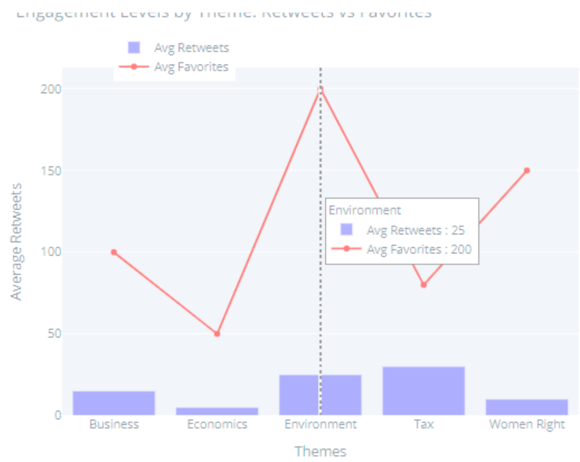
The second version used multiple line graphs to show changes in engagement across candidates in different moods, but the colour choices were not obvious enough, resulting in the graphs appearing cluttered and making it difficult to distinguish between different themes and mood variables. This was partly informed by an explanation of visual variables, in particular the considerations for colour use [7].



visualization 2

3) Failed version 3 (visualization 3):

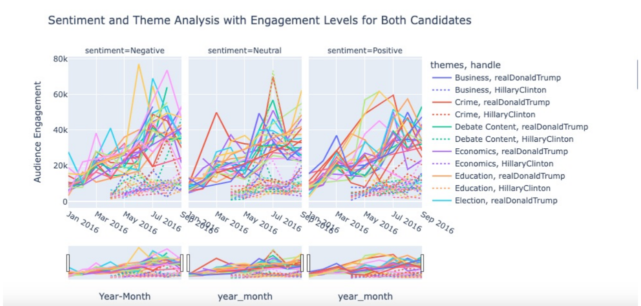
In the iteration a mix of bar and line graphs was utilized to illustrate the connection, between 'average retweets and 'likes. Although this method brought depth to the data presented the charts complexity and information overload made it challenging to grasp quickly. The ultimate resolution achieved an optimal blend of clarity and user friendly design [8].



visualization 3

C. Final adopted version (successful visualization 4):

The updated visualization now includes a line chart that effectively showcases the audience engagement levels based on various moods and themes of the candidates during the campaign duration. This new graph improves upon the version by providing a clearer depiction of mood shifts and theme performance through enhanced color selection and distinctive line styles.Grouped by year, month, candidate, topic and mood category and calculated average audience participation for each group. This design draws inspiration from the discussion, in Week05 regarding choosing chart types and creating a narrative with data [9].



visualization 4

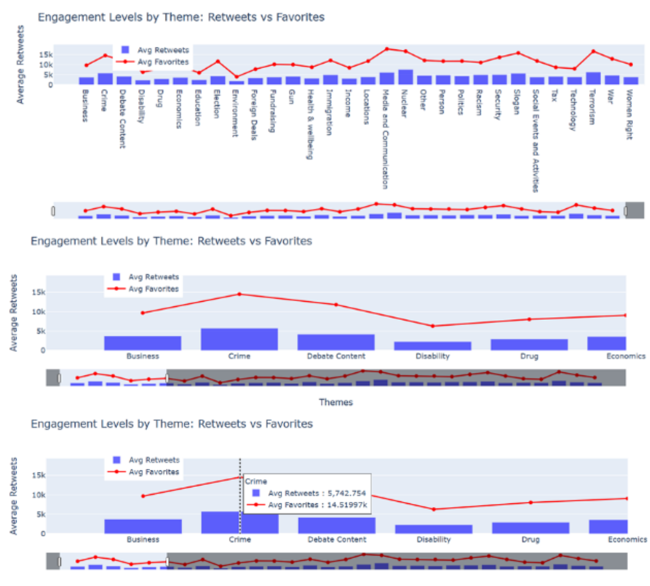
D. Improvement process towards a successful version:

1) Emphasis on theme and mood differentiation:

Different themes and moods are distinguished using a mix of colors and shapes in the design process—colors represent themes while shapes signify different emotions, like positivity or negativity or neutrality—allowing users to easily track emotional shifts and thematic variations intuitively[10].

2) Improved interaction design:

The interaction design combines a time slider and mouse hover functionality to allow users to focus on engagement with tweets over a certain period of time, and to get detailed information about various data points, such as topical concerns, sentiment analysis, and statistics on likes and retweets, as shown in Visualisations 5.[11].



visualization 5

3) Improvements in chart type selection:

The decision was made to use a line graph to display audience interaction and incorporate bar charts to enable viewers to observe patterns as well as concentrate, on particular information [12].

E. Identification and design of user interactions:

In creating this project I incorporated interactive elements such as sliders and zoom tools. These features allow users to navigate through the data dynamically by moving the mouse over areas to explore and extract insights, from varying viewpoints [13].

F. Code writing and implementation

I wrote code that created representations by plotting graphs with Matplotlib and incorporating labels and interactive features for enhanced user experience. I organized the data using the groupby() method. Determined the average audience engagement, for each group by utilizing the mean() function [14].

G. Design and analyse the assessment process:

I was involved in designing the evaluation process with a focus on how user interaction could improve data analysis methods.The goal was to make sure that the visualization accurately represented the data structure and highlighted variations, between themes [15].

REFERENCE

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[3] Week09-DR.pdf, "Relational Data and Aggregation Methods," University of Sydney.
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[5] Week06-EVAL.pdf, "Evaluation Methods and User-Centered Design," University of Sydney.
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